PatientTriage.ai

“Triage is a snapshot. Risk isn't.”

A Continuous Safety Decision-Support Layer for Emergency Waiting Rooms • Accenture Innovation Challenge 2026

COMPETITION TRACK	CORE STACK	DEPLOYMENT	REPOSITORY
Round 2 Technical Prototype	Python 3.13 / FastAPI / React 19	Edge / On-Premise Air-Gapped	github.com/freya1705/PatientTriageAI

1. Project Overview & Introduction

PatientTriage.ai is an emergency department clinical decision-support system and continuous physiological safety surveillance layer. While traditional emergency triage treats patient prioritization as a static, one-time snapshot at hospital intake, PatientTriage.ai continuously tracks patient waiting times, vital sign trajectory velocity (ΔSpO■, ΔHR), data uncertainty, and active clinical attention. The intelligence layer performs continuous physiological trend analysis, uncertainty scoring, confidence decay, and dynamic attention-gap prioritization, while deterministic safety rules provide hard guardrails and clinicians retain final decision authority.

The platform answers the single most critical operational question facing emergency clinicians: ***“Who in the waiting room is no longer safe to keep waiting?”***

2. The Core Problem & Competitive Moat (Vs. Legacy EHR Scores)

- **1. Silent Waiting Room Deterioration:** ESI Level 3/4 patients may worsen unmonitored; physiological decline may remain undetected until a subsequent reassessment or clinical deterioration becomes apparent.
- **2. Missing Vitals & Stale Data:** When vitals are missing, legacy systems default to low urgency. Under *Unknown is NOT Safe*, missing data heightens clinical vigilance.
- **3. The Attention Bottleneck:** Attended critical patients block static queues, while unattended deteriorating patients remain buried.
- **Competitive Moat vs. Native EHRs (Epic EDI / Cerner MEWS):** Legacy EHR algorithms are designed for admitted inpatients in hospital beds and only score raw severity. PatientTriage.ai is purpose-built for the waiting room and uniquely factors in **physician coverage** and **evidence staleness decay**.

3. System Architecture: 3-Tier Layered Design

Tier	Responsibilities & Scope	Key Modules
Tier 1: Deterministic Safety Layer	Deterministic red-flags (SpO■ < 85%, SBP < 75 mmHg, FAST Stroke, pediatric stridor) that bypass statistical models. Counterfactual downgrade safety blocking.	`safety_guardrails.py` `downgrade_guard.py`

Tier 2: AI & Decision Support	Continuous vital trajectory velocity (ΔSpO■, ΔHR), dynamic confidence decay ($\tau_{staleness}$), uncertainty scoring (Unknown ≠ Safe), Attention Gap re-ranking. Multimodal ingestion: BLE wearable oximetry rings/wristbands, kiosks, and nurse tablet walking rounds.	`risk_engine.py`
		`deterioration_engine.py`
		`attention_gap_engine.py`

Tier 3: Clinician Governance	Clinician override authority with mandatory justification recording, and immutable append-only audit logging.	`audit_service.py` `OverrideModal.jsx`

4. Core Intelligence Engines & Default Formula Weights

The Attention Gap Priority Equation:

```
Action Priority Score = (w_r × Risk + Urgency) + (w_d × Deterioration) + (w_s × Staleness) + Wait Hazard + (w_u × Uncertainty)
- (w_c × Clinical Coverage)
```

Default Parameter Bounds: w_r = 1.0 (base risk 0–100) • w_d = +25 to +40 pts (ΔSpO■ ≤ -5% or ΔHR ≥ +20 bpm) • w_s = +20 to +35 pts (safety window expiry) • w_u = +15 to +25 pts (missing vitals / zero history) • w_c = -35 pts (when is_attended = True, surfacing unattended deteriorating patients to Rank #1).

5. Benchmark Cohort & Empirical Impact Evaluation

Evaluated across 20 synthetic clinical scenarios representing 5 systematic emergency department failure modes:

Performance Dimension	Traditional Static Triage	PatientTriage.ai	Benchmark Impact
Waiting Deterioration Catch Rate	0/20 detected	**20/20 synthetic scenarios detected**	**100% Benchmark Coverage**
Stale Observation Flagging	0/20 flagged	**20/20 synthetic cases flagged (EXPIRED)**	**Zero Unmonitored Stale Waits**
False Reassurance on Missing Vitals	High (Treated Normal)	**0% False Reassurance (Unknown ≠ Safe)**	**Eliminates Under-Triage**
Attention Gap Optimization	None (Attended Block)	**Active (Elevates Unattended)**	**Optimized Clinician Utilization**
Unsafe Priority Downgrades Blocked	0 Guardrails	**100% Guarded (Proof Required)**	**100% Downgrade Guarded**

**Note: Results reflect simulated evaluations across 20 synthetic clinical benchmark scenarios for prototype demonstration.*

6. Phased Implementation Roadmap & Precise Trial Endpoints

- **Phase 1 (Q3 2026):** Lab Validation • 20 synthetic scenarios verified across 33 automated tests; sub-15ms latency.
- **Phase 2 (Q4 2026):** Shadow Clinical Trial • Silent background deployment with Epic/Cerner via HL7 FHIR and CDS Hooks.
- **Phase 3 (Q1–Q2 2027):** Live Pilot • **Primary Safety Endpoint:** Reduction in Mean Time to Escalation (MTTE) (>45% faster); **Operational Endpoint:** Nurse false-alarm rate strictly < 2 alerts/nurse/shift; **Economic Endpoint:** ≥25% reduction in Left-Without-Being-Seen (LWBS).
- **Phase 4 (Q3 2027+):** Multi-Hospital Enterprise Network Scaling • Regional dashboarding and telemedicine integration.

7. Quick Start & Security Architecture

```
# PowerShell 1-Command Launch: .\start.ps1 | Run 33 Automated Tests: python -m pytest -v
```

- **Zero PHI:** 100% synthetic physiological datasets • **Air-Gapped:** Zero third-party cloud LLM calls • **Audit Ledger:** Append-only event logs.

Regulatory & Safety Notice: PatientTriage.ai is a clinical decision-support research prototype developed for the Accenture Innovation Challenge 2026. All patient cohorts are synthetically generated. This system is not a certified medical device and does not replace professional clinical judgment.